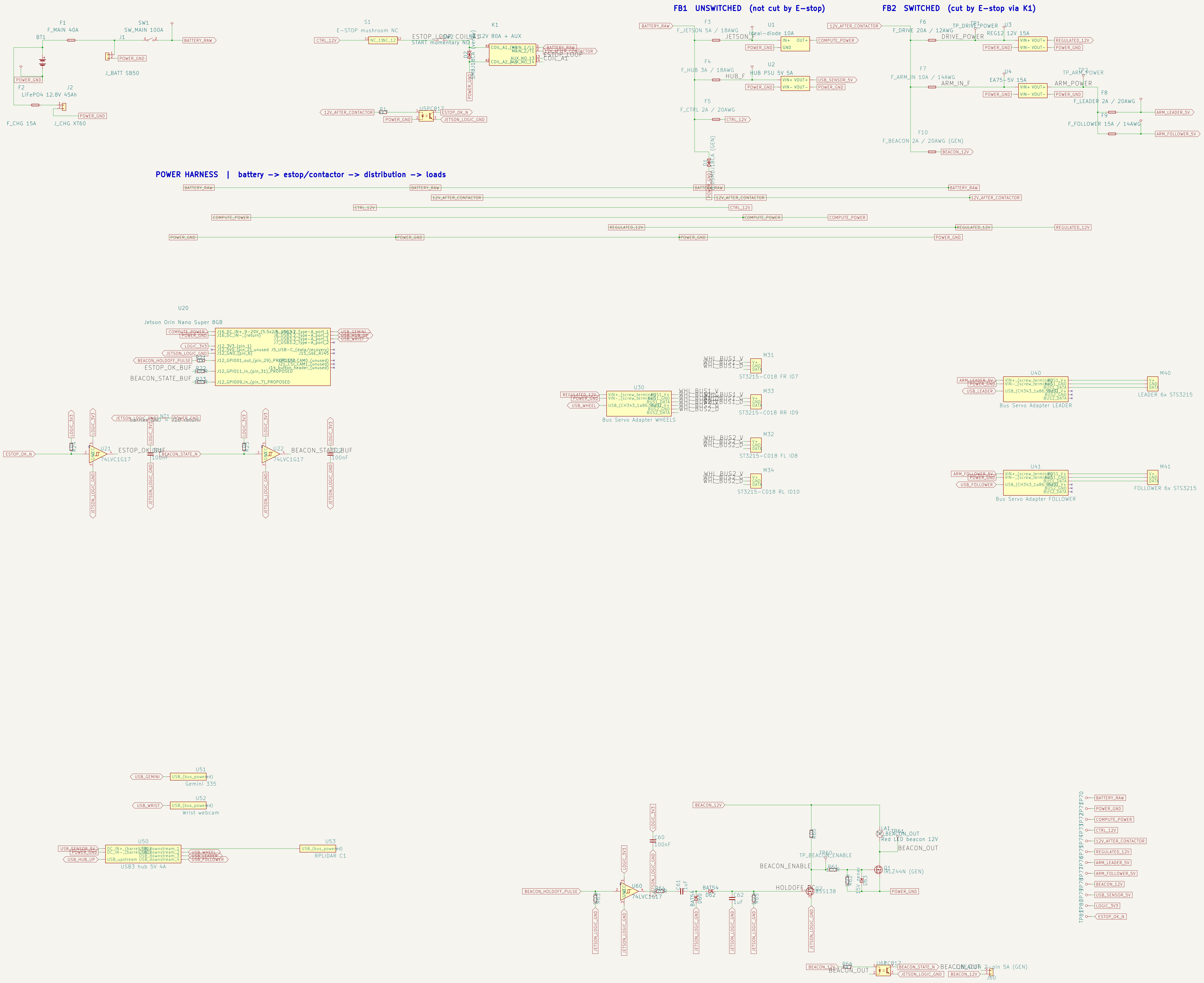
KIRO Wiring V2 — Battery Power System


FB1 UNSWITCHED (not cut by E-stop)
FB2 SWITCHED (cut by E-stop via K1)
POWER HARNESS | battery -> estop/contactor -> distribution -> loads
BT1
LiFePO4 12.8V 45Ah
F1
F_MAIN 40A
F2
F_CHG 15A
J1
J_BATT SB50
J2
J_CHG XT60
SW1
SW_MAIN 100A
BATTERY_RAW
POWER_GND
S1
E-STOP mushroom NC
NC_11
NC_12
CTRL_12V
ESTOP_LOOP
SW2
START momentary NO
COIL_A1
K1
12V 80A + AUX
COIL_A1_(+)
COIL_A2_(-)
MAIN_1/L1
MAIN_2/T1
AUX_NO_13
AUX_NO_14
12V_AFTER_CONTACTOR
D2
SMBJ18CA (verify)
R1
4k7
U5PC817
ESTOP_OK_N
JETSON_LOGIC_GND
F3
F_JETSON 5A / 18AWG
JETSON_F
U1
Ideal-diode 10A
IN+
OUT+
GND
COMPUTE_POWER
F4
F_HUB 3A / 18AWG
HUB_F
U2
HUB PSU 5V 5A
VIN+
VOUT+
VIN-
VOUT-
USB_SENSOR_5V
F5
F_CTRL 2A / 20AWG
D1
SMDJ18CA (GEN)
F6
F_DRIVE 20A / 12AWG
DRIVE_POWER
TP1
TP_DRIVE_POWER
U3
REG12 12V 15A
REGULATED_12V
F7
F_ARM_IN 10A / 14AWG
ARM_IN_F
U4
EA75-5V 15A
TP2
TP_ARM_POWER
ARM_POWER
F8
F_LEADER 2A / 20AWG
ARM_LEADER_5V
F9
F_FOLLOWER 15A / 14AWG
ARM_FOLLOWER_5V
F10
F_BEACON 2A / 20AWG (GEN)
BEACON_12V
U20
Jetson Orin Nano Super 8GB
J16_DC_IN+_9-20V_(5.5x2.5_barrel)
J16_DC_IN-_(return)
J12_3V3_(pin_1)
J12_5V0_(pin_2)_unused
J12_GND_(pin_6)
J12_GPIO01_out_(pin_29)_PROPOSED
J12_GPIO11_in_(pin_31)_PROPOSED
J12_GPIO09_in_(pin_7)_PROPOSED
J6_USB3.2_Type-A_port_1
J6_USB3.2_Type-A_port_2
J7_USB3.2_Type-A_port_1
J7_USB3.2_Type-A_port_2
J5_USB-C_(data/recovery)
J15_GbE_RJ45
J20_CSI_CAM0_(unused)
J21_CSI_CAM1_(unused)
J14_button_header_(unused)
USB_GEMINI
USB_HUB_UP
USB_WRIST
LOGIC_3V3
BEACON_HOLDOFF_PULSE
R21
1k
ESTOP_OK_BUF
R22
100R
BEACON_STATE_BUF
R23
100R
NT1
carrier GND = J16 return
R24
10k
U21
74LVC1G17
C21
100nF
BEACON_STATE_N
R25
10k
U22
74LVC1G17
C22
100nF
U30
Bus Servo Adapter WHEELS
VIN+_(screw_terminal)
VIN-_(screw_terminal)
USB_(CH343_1a86:55d3)
BUS1_V+
BUS1_GND
BUS1_DATA
BUS2_V+
BUS2_GND
BUS2_DATA
USB_WHEEL
WHL_BUS1_V
WHL_BUS1_G
WHL_BUS1_D
WHL_BUS2_V
WHL_BUS2_G
WHL_BUS2_D
M31
ST3215-C018 FR ID7
M33
ST3215-C018 RR ID9
M32
ST3215-C018 FL ID8
M34
ST3215-C018 RL ID10
V+
GND
DATA
U40
Bus Servo Adapter LEADER
USB_LEADER
M40
LEADER 6x STS3215
U41
Bus Servo Adapter FOLLOWER
USB_FOLLOWER
M41
FOLLOWER 6x STS3215
U51
USB_(bus_powered)
Gemini 335
U52
Wrist webcam
U50
USB3 hub 5V 4A
DC_IN+_(barrel)
DC_IN-_(barrel)
USB_upstream
USB_downstream_1
USB_downstream_2
USB_downstream_3
USB_downstream_4
U53
RPLIDAR C1
U60
74LVC1G17
R63
100k
C60
100nF
R64
220R
C61
1uF
D60
BAT54
D62
BAT54
C62
1uF
R65
100k
HOLDOFF
Q2
BSS138
R60
BEACON_ENABLE
TP60
TP_BEACON_ENABLE
R61
100R
R62
100k
D63
15V zener
Q1
IRLZ44N (GEN)
LA1
Red LED beacon 12V
TP61
TP_BEACON_OUT
BEACON_OUT
R66
4k7
U61PC817
J60
BEACON 2-pin 5A (GEN)
TP70
TP71
TP72
TP73
TP74
TP75
TP76
TP77
TP78
TP79
TP80
TP81

Flattened from hierarchical sheets for single-file delivery
Genie EduTech
Sheet: /
File: kiro_wiring_v2.kicad_sch
Title: KIRO Wiring V2 (flat)
Size: A0 Date: 2026-11-12 **Rev: flat-1**
KiCad E.D.A. 10.0.7 Id: 1/1